

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

TRAINING TR-102 REPORT DAY 8

4 August 2025

Overview:

We covered a wide range of important Python programming concepts. These included working with strings, performing slicing and concatenation, understanding different types of operators, and using data structures like list, tuple, dictionary, and set. We also practiced using loops and creating functions to organize code better. To apply all these concepts, we created a simple calculator program using Python. It was a fun and practical session that helped us connect many basics together.

Learning Objectives:

- Learn how to work with strings and perform slicing & joining.
- Understand different data structures and their uses.
- Use loops and functions to make the code efficient.
- Build a basic calculator using Python.

Python String:

Strings in python are surrounded by either single quotation marks, or double quotation marks.

'hello' is the same as "hello".

We can display a string literal with the print() function:

Example

```
print("Hello")  
print('Hello')
```

Output:

```
Hello  
Hello
```

Python Slicing String:

It can return a range of characters by using the slice syntax. It specifies the start index and the end index, separated by a colon, to return a part of the string.

Example:

```
b = "Hello, World!"  
print(b[2:5])
```

Output:

```
llo
```

Python Concatenation String:

To concatenate, or combine, two strings you can use the + operator.

Example:

```
a = "Hello"  
b = "World"  
c = a + b  
print(c)
```

Output:

```
HelloWorld
```

Python Operators:

Operators in general are used to perform operations on values and variables. These are standard symbols used for logical and arithmetic operations. There are two different types of **Python operators**.

- **OPERATORS:** These are the special symbols. Eg- + , * , /, etc.
- **OPERAND:** It is the value on which the operator is applied.

Types of Operators in Python

Operators in Python

Operators	Type
+, -, *, /, %	Arithmetic operator
<, <=, >, >=, ==, !=	Relational operator
AND, OR, NOT	Logical operator
&, , <<, >>, ~, ^	Bitwise operator
=, +=, -=, *=, %=	Assignment operator

Loops in Python:

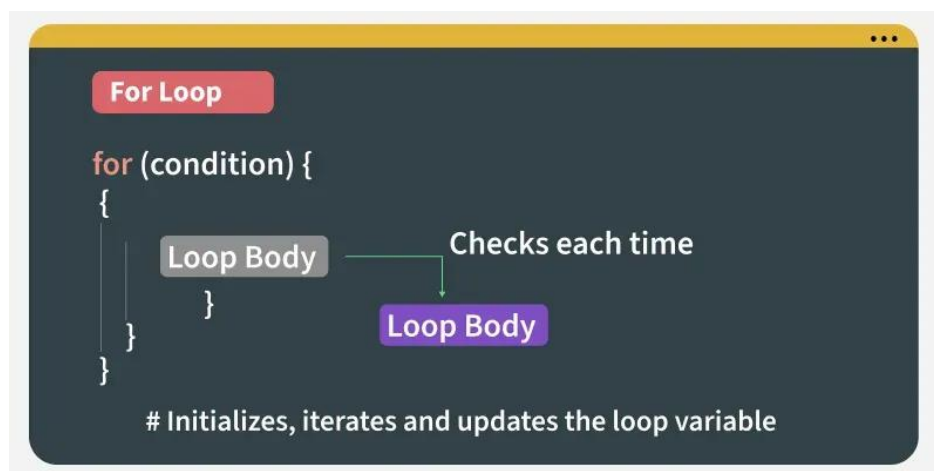
Loops in Python are used to repeat actions efficiently. The main types are For loops (counting through items) and While loops (based on conditions).

For Loop in Python

For loops is used to iterate over a sequence such as a list, tuple, string or range. It allows to execute a block of code repeatedly, once for each item in the sequence.

Syntax:

```
for          iterator_var          in          sequence:
statements(s)
```



Example:

```
n = 4
for i in range(0, n):
    print(i)
```

Output:

```
0
1
2
3
```

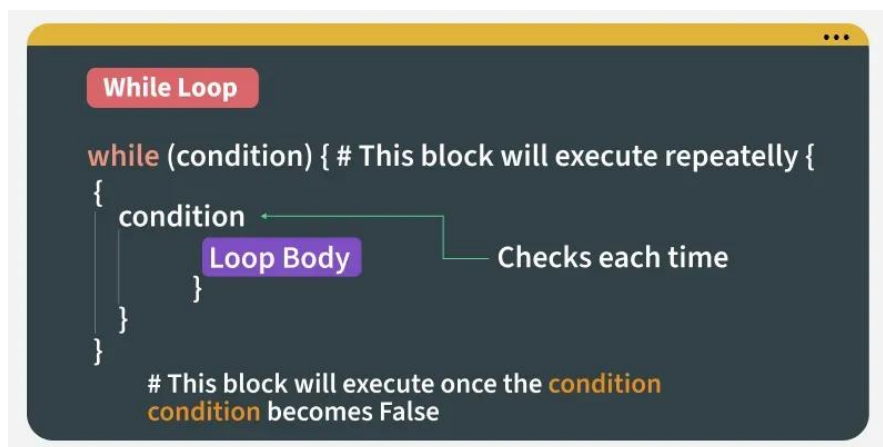
While Loop in Python

In Python, a while loop is used to execute a block of statements repeatedly until a given condition is satisfied. When the condition becomes false, the line immediately after the loop in the program is executed.

Syntax:

```
while
statement(s)
```

expression:



Example:

```
cnt = 0
while (cnt < 3):
```

```
cnt = cnt + 1  
print("Hello World")
```

Output:

```
Hello World  
Hello World  
Hello World
```

Python Functions

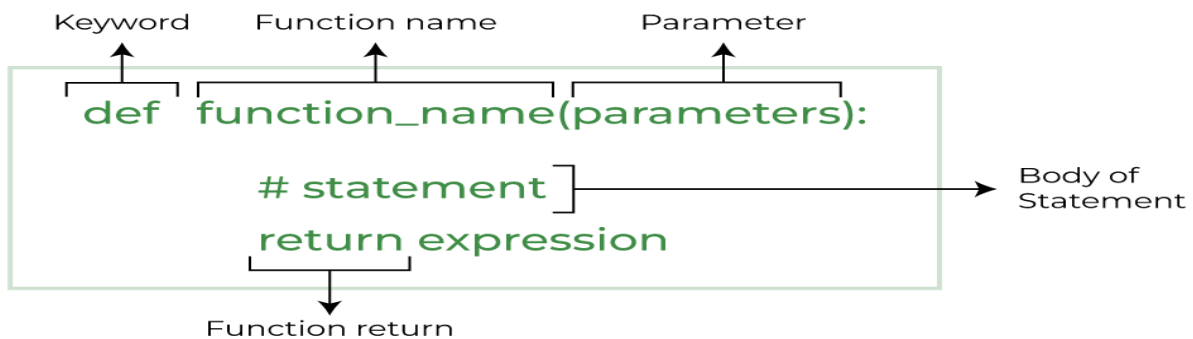
Python Functions is a block of statements that does a specific task. The idea is to put some commonly or repeatedly done task together and make a function so that instead of writing the same code again and again for different inputs,

Benefits of Using Functions:

- Code Reuse
- Reduced code length
- Increased readability of code

Python Function Declaration

The syntax to declare a function is:



Types of Functions in Python

The different types of functions in Python are:

- **Built-in library function:** These are Standard functions in Python that are available to use.
- **User-defined function:** We can create our own functions based on our requirements.

Defining a Function:

```
def fun():  
    print("Welcome to GFG")
```

Calling a Function

```
def fun():  
    print("Welcome to GFG")  
  
# Driver code to call a function  
fun()
```

Calculator Using Python :

We created a simple calculator program that can perform basic arithmetic operations such as addition, subtraction, multiplication, and division. It uses functions to keep the code organized and if-elif-else statements to handle the operation the user chooses.

Concepts Used:

- **Functions** (def)
- **User input** (input())
- **Type casting** (int() or float())
- **Conditionals** (if-elif-else)
- **Loop** (optional, if we want to repeat operations)

Conclusion:

The session was very productive and hands-on. It helped in combining many concepts and understanding how they work together. Creating the calculator program made the learning fun and practical. These basics are very important for building real applications and practicing them boosted confidence in writing better Python code.